

Variable Selection using Artificial Neural Networks

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Abstract

Variable selection in neural networks remains challenging due to their nonlinear and highly parameterized structure, in particular because first-layer weights cannot be interpreted independently of the downstream network. Yet, selecting a relevant subset of input variables can improve interpretability while reducing model size. We introduce an adaptive first-layer regularization framework for multilayer perceptrons relying on a single regularization parameter. Each first-layer neuron is weighted by its downstream sensitivity, yielding a scale-invariant penalty and normalizing the corresponding first-layer gradients. The penalized problem is solved on the representatives of the rescaling symmetry where every sensitivity equals one, where it becomes a lasso on the first layer, by a proximal adaptive gradient method whose zero-thresholding test is the calibrated one. To preserve variables whose relevance emerges only through nonlinear interactions, the regularization level is raised progressively along a warm-started path. The resulting sparse networks discard irrelevant input variables and their associated parameters, thereby reducing both input dimensionality and model complexity. We develop the framework for Gaussian regression, binary classification, and Cox proportional-hazards models, and investigate its variable-selection and sparsification properties through numerical experiments.

Keywords: neural networks, variable selection, sparse neural networks, adaptive regularization, pivotal loss; proximal optimization

1 Introduction

In machine learning, models ranging from linear methods to artificial neural networks (ANNs) have achieved remarkable success: the former owing largely to their simplicity and interpretability, and the latter to their flexibility and ability to learn complex relationships from data. Selecting which features a model should be allowed to use, however, is a statistical problem before it is a computational one. In the spirit of Occam’s razor—“*Entities must not be multiplied beyond necessity*”—among models with comparable predictive performance, preference should be given to those that achieve it with fewer parameters. Yet modern machine learning has increasingly favoured predictive performance and large-scale parametrization over parsimony. As computational power and storage have become more readily available, it has become feasible to train increasingly large models using many inputs and parameters, often without explicitly questioning whether all of them are necessary. In this regime, overparameterization is computationally affordable, but computational feasibility does not imply statistical necessity. A neural network may therefore achieve strong predictive performance while relying on an unnecessarily large set of inputs and parameters, making it difficult to distinguish the variables that genuinely contribute to the model from those that are merely accommodated by its flexibility. As a result, the input-layer weight

matrix W_1 may remain unnecessarily dense: irrelevant input variables retain connections to the hidden layer even though these connections are not needed for prediction. Identifying and removing such variables therefore not only improves interpretability but can also substantially reduce the number of parameters required by the network.

For linear models, this variable-selection problem is well posed and extensively studied. The lasso of Tibshirani (1996), for instance, produces a family of supports indexed by a regularization parameter λ , and a substantial literature addresses how this parameter should be chosen. In neural networks, the same question arises in a fundamentally different setting: the parameterization is non-identifiable, the training objective is non-convex, and the fitted model is typically obtained through first-order iterative optimization rather than by solving the optimization problem to global optimality. Sparsity-inducing penalties on the input layer extend the principle of variable selection to this setting (Feng and Simon, 2019; Lemhadri et al., 2021), but the crucial choice of λ is still most commonly delegated to cross-validation.

Cross-validation is unattractive for variable selection for two reasons. First, it is computationally expensive, as it multiplies the cost of an already demanding training procedure by the number of folds and candidate values of λ . More fundamentally, it targets predictive risk rather than variable-selection accuracy: the value of λ that minimizes prediction error need not recover the underlying support and may retain variables carrying little or no signal. If the objective is variable selection, the regularization level should instead be calibrated directly against the event one seeks to control—the inclusion of variables carrying no predictive signal. A natural approach is therefore to calibrate λ under the pure-noise model, in which no variable carries predictive signal, so that, for a prescribed α , no variable is selected with probability at least $1 - \alpha$. Such null-based calibration has been developed for Gaussian linear regression (Donoho and Johnstone, 1994; Belloni et al., 2011) and, more recently, extended to a broader class of loss functions through the pivotal information criterion of Sardy et al. (2026). The pivotal construction makes the null distribution of the calibrating statistic independent of unknown nuisance parameters, allowing λ to be calibrated without estimating quantities such as the noise scale. In this work, we adopt this pivotal framework and the corresponding loss functions considered in Sardy et al. (2026).

Transferring this calibration principle to neural networks introduces an additional difficulty. The gradient with respect to the input-layer weights depends on the downstream network weights, which, because neural-network parameterizations are non-identifiable, can be rescaled without changing the function represented by the network while simultaneously changing the magnitude of the input-layer gradients. Consequently, a regularization threshold based directly on these gradients is itself parameterization-dependent: the same fitted function may cross or remain below the threshold solely as a consequence of rescaling. A meaningful null calibration for neural networks must therefore account for the arbitrary scale induced by the network parameterization.

This paper addresses this difficulty by exploiting the structure of the input-layer gradient under the null. We show that, when the input layer is empty, this gradient is exactly a rank-one matrix: the outer product of a network-sensitivity vector and a data-dependent vector. The data-dependent factor carries the evidence against the null and is calibrated once, before training, by Monte Carlo, whereas the sensitivity factor captures the current scale of the network and is recomputed at every iteration. This decomposition yields an

adaptive regularization level that follows the scale of the network throughout training, making variable selection invariant to equivalent rescalings of its parameterization while preserving the original null calibration. Following the pivotal framework described above, we focus on three tasks: regression, binary classification, and Cox survival analysis.

Contributions.

- A zero-thresholding condition for a penalized multilayer perceptron (Theorem 2) and an exact rank-one factorization of the null gradient (proof of Theorem 2) separating the statistical evidence from the network amplification.
- A regularization rule, $\lambda^{(t)} = \lambda_{\alpha}^{\text{DB}} s(\boldsymbol{\theta}^{(t)})$, in which only the first factor is calibrated. It is exactly invariant under the rescaling group of the network (Proposition 1), recovers the fixed-architecture condition at the null (Theorem ??) uniformly over the unpenalized parameters, and is the largest admissible such rule (Proposition ??).
- The consequence that the calibrated threshold does not depend on the architecture (Corollary ??): one calibration serves every depth and width, and collapses to the linear-model threshold.
- Pivotal losses and closed-form null gradients for Gaussian regression, binary classification and the Cox model (Table 1), reducing the calibration to a single matrix product per Monte Carlo draw.
- An optimization scheme (Section 5) that keeps every sensitivity equal to one, where the penalty becomes a plain lasso on the first layer, and takes proximal adaptive steps along a warm path in λ . At the null, it performs exactly the calibrated zero-thresholding test (Proposition 4).

2 Related Work

Sparse-input networks. Penalizing the input layer of a network to obtain feature selection is an established device. Feng and Simon (2019) fit sparse-input networks with a group penalty on the input weights, Louizos et al. (2018) relax an ℓ_0 count with stochastic gates, as do Yamada et al. (2020), and Lemhadri et al. (2021) tie the input layer to a residual linear path so that the selected support is nested along the path. In all of these the regularization level is a tuning parameter, chosen by cross-validation or by a target sparsity; the present paper leaves the architecture and the penalty alone and supplies the level.

Selection with error control. A second line of work attaches a frequentist guarantee directly to the selected set. Knockoffs (Barber and Candès, 2015) and their model-X extension (Candès et al., 2018) control the false discovery rate by comparing each original variable with an artificial counterpart designed to mimic the dependence structure of the covariates while carrying no additional information about the response. DeepPINK (Lu et al., 2018) extends this principle to neural networks through a dedicated architecture that jointly processes original variables and their knockoffs. These approaches provide a stronger inferential guarantee than the one pursued here, but require the construction of

valid knockoff variables, which in the model-X setting relies on knowledge or accurate estimation of the covariate distribution. Our approach instead operates directly on the original covariates and can be integrated into a standard MLP without generating artificial variables or modifying the architecture to accommodate them. The resulting guarantee is different in nature: our calibration controls the zero-thresholding behavior at the null model rather than the false discovery rate of the returned support.

Tuning without cross-validation. Choosing λ from the null rather than from held-out risk goes back to the universal threshold of Donoho and Johnstone (1994) and, for the lasso, to the theoretical choice of Bickel et al. (2009). The quantile universal threshold (Giacobino et al., 2017) replaces the asymptotic rate by a finite-sample quantile of the zero-thresholding statistic, and the pivotal information criterion (Sardy et al., 2026) makes that quantile computable by removing the nuisance parameters from the statistic. This paper is the extension of that programme to a nonlinear, non-identified model.

Rescaling invariance. That a positively homogeneous network can move mass between layers at no cost, so that a norm read off one layer means nothing on its own, is the phenomenon behind path-normalized measures of capacity (Neyshabur et al., 2015b) and behind Path-SGD (Neyshabur et al., 2015a), which replaces gradient descent by a descent invariant to these rescalings. We take a different route, dictated by what the calibration needs: we keep the ℓ_1 penalty on $W^{(1)}$ and rescale the *level*, by exactly the factor that appears in the null gradient. The optimization then moves, at every step, to the rescaled network where this factor equals one; moves of this kind, which leave the function unchanged, are also used by equi-normalization (Stock et al., 2019) and symmetry teleportation (Zhao et al., 2022). The closest algorithm is PathProx (Yang et al., 2022), which normalizes the input weights of every ReLU neuron and takes proximal steps on the output weights; here the normalizing factor depends on the data and on the deeper layers, which adds one term to the gradient (Section 5).

3 Sparse Learning with Neural Networks

3.1 Setting and Notation

Let $\{(\mathbf{x}_i, y_i)\}_{i=1}^n$ be independent observations, with $\mathbf{x}_i \in \mathbb{R}^p$, and let $X \in \mathbb{R}^{n \times p}$ denote the design matrix. Throughout, without loss of generality, we normalize

$$\frac{1}{n} \sum_{i=1}^n x_{ij}^2 = 1 \quad (j = 1, \dots, p). \quad (1)$$

Consider a fully connected artificial neural network (ANN), also known as a multilayer perceptron (MLP), with L layers, defining a class of functions

$$f_{\boldsymbol{\theta}}(\mathbf{x}) = S_L \circ S_{L-1} \circ \dots \circ S_1(\mathbf{x}).$$

Throughout this work, we restrict attention to MLPs with a scalar output, so that $f_{\boldsymbol{\theta}} : \mathbb{R}^p \rightarrow \mathbb{R}$. The parameter vector $\boldsymbol{\theta}$ collects all model parameters. For observation i , we write $\eta_i = f_{\boldsymbol{\theta}}(\mathbf{x}_i)$, and $\boldsymbol{\eta} = f_{\boldsymbol{\theta}}(X) \in \mathbb{R}^n$ for the vector of outputs over the n observations. Write $p_0 = p$ for the input dimension and p_l for the number of neurons in layer l , so

that $W^{(l)} \in \mathbb{R}^{p_l \times p_{l-1}}$ and $\mathbf{b}^{(l)} \in \mathbb{R}^{p_l}$ denote the weight matrix and bias of layer l . For $l \in \{1, \dots, L-1\}$, layer l applies an affine map followed by a componentwise nonlinearity,

$$S_l(\mathbf{u}) = \sigma_l(\mathbf{b}^{(l)} + W^{(l)}\mathbf{u}),$$

mapping the p_{l-1} -vector $\mathbf{u}$ into a p_l -vector. The output layer is purely affine,

$$S_L(\mathbf{u}) = \mathbf{b}^{(L)} + W^{(L)}\mathbf{u}, \quad p_L = 1,$$

with no activation applied: any transformation of the output, such as a sigmoid for binary classification, is absorbed into the loss so that regression ($y \in \mathbb{R}$), binary classification ($y \in \{0, 1\}$) and survival data are handled by the same architecture.

With an MLP, the relevant features are the columns of the first-layer weight matrix $W^{(1)}$ that are non-zero, as these are the only ones directly connected to the input X . For this identification to be meaningful, a small entry of $W^{(1)}$ must not be reproducible by inflating weights in deeper layers: otherwise an input could influence the output while its first-layer column is arbitrarily small, severing the link between $W^{(1)}$ and feature relevance and nullifying any penalty applied to $W^{(1)}$.

The quantity that governs this amplification is the sensitivity of the output to the first layer. Write $\mathbf{h}_i^{(1)} = \mathbf{b}^{(1)} + W^{(1)}\mathbf{x}_i$ for the first-layer preactivation at observation i and define the *sensitivity vectors*

$$\mathbf{a}_i(\boldsymbol{\theta}) = \frac{\partial \eta_i}{\partial \mathbf{h}_i^{(1)}} = D_i^{(1)} W^{(2)\top} \dots D_i^{(L-1)} W^{(L)\top} \in \mathbb{R}^{p_1}, \quad i = 1, \dots, n. \quad (2)$$

where $D_i^{(\ell)} = \text{diag}(\sigma'_\ell(\mathbf{h}_i^{(\ell)}))$. They measure how strongly the network output responds to each first-layer unit, and they are the exact factor by which a gradient signal is amplified between the penalized weights and the output. For each first-layer neuron k , its *neuron sensitivity scale* is defined by

$$s_k(\boldsymbol{\theta}) = \|\mathbf{a}_{\cdot k}(\boldsymbol{\theta})\|_\infty = \max_{1 \leq i \leq n} |a_{ik}(\boldsymbol{\theta})|. \quad (3)$$

Section 3.2 shows that measuring the penalty on this sensitivity scale removes the arbitrary neuron-wise scaling of the network.

Throughout this work, we use the leaky ReLU activation

$$\sigma_\ell(t) = \max\{t, mt\}, \quad \ell = 1, \dots, L-1, \quad m \in (0, 1),$$

in all hidden layers. This choice is motivated by two properties that are particularly useful for the proposed regularization.

First, the derivative of each hidden-layer activation remains bounded away from zero wherever it is defined. More precisely,

$$|\sigma'_\ell(t)| \geq m > 0,$$

This prevents the activation functions themselves from completely blocking the propagation of information through the network and, in particular, the backpropagated signal used

to construct the sensitivity vectors $\mathbf{a}_i$. This property is also relevant to the structural interpretation of variable selection through $W^{(1)}$. If an intermediate activation contains a flat region, an input–output path may be blocked by the activation even though the corresponding entries of $W^{(1)}$ are nonzero. The network may then become functionally independent of some input variables without the associated first-layer weights being zero.

Second, the leaky ReLU is positively homogeneous:

$$\sigma_\ell(ct) = c \sigma_\ell(t), \quad c > 0.$$

Although this property holds for every hidden layer, it is specifically required at the first one for the scale-invariance of the adaptive penalty developed in Section 3.2.

3.2 The Scale Ambiguity and the Adaptive Regularization

A multilayer network does not uniquely identify its own parameters, and this ambiguity affects precisely the quantity that a penalty on $W^{(1)}$ is meant to measure. By positive homogeneity of the first-layer activation, for any first-layer neuron k and any $c > 0$, the reparameterization

$$W_{k,\cdot}^{(1)} \mapsto c W_{k,\cdot}^{(1)}, \quad b_k^{(1)} \mapsto c b_k^{(1)}, \quad W_{\cdot,k}^{(2)} \mapsto \frac{1}{c} W_{\cdot,k}^{(2)} \quad (4)$$

leaves the fitted function f_θ , and therefore the loss, unchanged, while multiplying $\|W_{k,\cdot}^{(1)}\|_1$ by c . Read in the direction $c \rightarrow 0$, this is the failure mode announced in Section 3.1: the network drives its input weights towards zero, compensates by inflating $W^{(2)}$, predicts exactly what it predicted before, and pays a penalty as small as it likes.

The sensitivity scale (3) transforms in the opposite direction. Indeed, under (4),

$$\mathbf{a}_{\cdot,k}(\theta) \mapsto \frac{1}{c} \mathbf{a}_{\cdot,k}(\theta), \quad s_k(\theta) \mapsto \frac{1}{c} s_k(\theta).$$

The sensitivity scale therefore provides exactly the compensating factor needed to remove this arbitrary parameter scaling.

Proposition 1 *The adaptive first-layer penalty*

$$\mathcal{P}(\theta) = \sum_{k=1}^{p_1} s_k(\theta) \|W_{k,\cdot}^{(1)}\|_1 \quad (5)$$

is invariant under any positive neuron-wise reparameterization of the form (4). Consequently, any penalized criterion does not depend on the arbitrary scale chosen between a first-layer neuron and the following layer.

The role of the sensitivity scale extends beyond scale invariance. By matching the sensitivity factor appearing in the first-layer gradient, it removes neuron-specific downstream amplification from the regularization threshold, thereby enabling the calibration of a single regularization parameter λ , as developed in Section 4.

Since the rescaling (4) multiplies s_k by $1/c$, choosing $c = s_k(\theta)$ for every neuron with $s_k(\theta) > 0$ gives an equivalent network, computing the same function with the same penalty, on which every sensitivity equals one. We call it the *normalized* network.

3.3 The Learning Problem

Given data $(X, \mathbf{y})$, we consider the training problem

$$\min_{\boldsymbol{\theta} \in \mathbb{R}^\gamma} J_\lambda(\boldsymbol{\theta}), \quad J_\lambda(\boldsymbol{\theta}) = \ell(\boldsymbol{\theta}) + \lambda \sum_{k=1}^{p_1} s_k(\boldsymbol{\theta}) \|W_{k,\cdot}^{(1)}\|_1, \quad \ell(\boldsymbol{\theta}) = \mathcal{L}_n(f_{\boldsymbol{\theta}}(X); \mathbf{y}), \quad (\text{P})$$

where $\mathcal{L}_n(\boldsymbol{\eta}; \mathbf{y})$ is a data-dependent goodness-of-fit measure of the vector of outputs, and $\lambda \geq 0$ is the regularization parameter controlling the strength of the adaptive first-layer penalty, set to the level $\lambda_\alpha^{\text{DB}}$ developed in Section 4. The full expression J_λ is referred to as the cost function. Throughout this work, we consider three statistical learning settings: regression, binary classification, and survival analysis under the Cox proportional hazards model. In each case, the goodness-of-fit term $\mathcal{L}_n$ is chosen with a specific normalization that will play a central role in the calibration of λ . We consider the same three losses as Sardy et al. (2026).

For regression, with $\mathbf{y} \in \mathbb{R}^n$, we use the square-root loss

$$\mathcal{L}_n^{\text{reg}}(\boldsymbol{\eta}; \mathbf{y}) = \frac{1}{\sqrt{n}} \|\mathbf{y} - \boldsymbol{\eta}\|_2.$$

For binary classification, with $y_i \in \{0, 1\}$ and $\mu_i = 1/(1 + e^{-\eta_i})$ denoting the logit produced by the network, we use the transformed logistic loss

$$\mathcal{L}_n^{\text{log}}(\boldsymbol{\eta}; \mathbf{y}) = \frac{2}{n} \sum_{i=1}^n \left[y_i \sqrt{\frac{1 - \mu_i}{\mu_i}} + (1 - y_i) \sqrt{\frac{\mu_i}{1 - \mu_i}} \right].$$

For survival analysis, let (y_i, δ_i) denote the observed time and event indicator for subject i . We use the square-root negative Cox partial log-likelihood

$$\mathcal{L}_n^{\text{Cox}}(\boldsymbol{\eta}; \mathbf{T}, \boldsymbol{\delta}) = \sqrt{\sum_{i=1}^n \delta_i \left[\eta_i - \log \left(\sum_{j: y_j \geq y_i} e^{\eta_j} \right) \right]}.$$

These particular forms and normalizations are not merely conventional choices. They are selected so that the score of the loss at the null model is a pivotal quantity, free of any nuisance parameters. This property is what ultimately allows our choice of regularization level $\lambda_\alpha^{\text{DB}}$ developed in Section 4.

The numerical solution of (P) is considered separately in Section 5. In particular, we discuss why the final regularization level should be approached along a regularization path rather than imposed directly at initialization, and describe the optimization scheme used to solve the successive penalized problems.

3.4 The Problem Actually Solved

Since the rescalings change neither the fit nor the penalty, (P) can be solved on normalized networks only. There the penalty is a plain lasso on the first layer:

$$\min_{\boldsymbol{\theta}} \ell(\boldsymbol{\theta}) + \lambda \|W^{(1)}\|_1 \quad \text{subject to} \quad s_k(\boldsymbol{\theta}) = 1, \quad k = 1, \dots, p_1. \quad (\text{C})$$

Problems (P) and (C) have the same optimal value, and normalizing a solution of the first gives a solution of the second. The selected set $\hat{S} = \{j : \hat{W}_{\cdot j}^{(1)} \neq \mathbf{0}\}$ does not depend on the rescaling, which never turns a zero weight into a non-zero one. The optimization of Section 5 works on (C).

4 Choice of λ

4.1 Calibration under the null

The regularization parameter λ in (P) governs the sparsity of the learned model: if λ is too large, the penalty dominates the loss and drives $W^{(1)} = \mathbf{0}$, thereby discarding all input features; if λ is too small, irrelevant features may enter the model. A standard approach is to select λ by cross-validation. This is unattractive in the present setting for two reasons. First, repeated fitting is computationally expensive for neural networks. Second, cross-validation targets predictive performance rather than support recovery and may therefore favour insufficient regularization when the objective is variable selection.

We instead build on the null-calibration principle of Sardy et al. (2026). The central idea is to calibrate a reference regularization level under the pure-noise model $H_0 : W^{(1)} = \mathbf{0}$ so that, with prescribed probability $1 - \alpha$, the penalized criterion (P) admits $W^{(1)} = \mathbf{0}$ as a local minimum.

In a multilayer model, the gradient with respect to $W^{(1)}$ depends not only on the statistical association between X and $\mathbf{y}$, but also on how strongly the downstream network propagates perturbations of the first layer to the output. These two contributions play fundamentally different roles. The former is determined by the data and can be calibrated under H_0 , whereas the latter depends on the parametrization of the downstream network and may evolve during training. However, the adaptive first-layer penalty (5) is constructed using precisely the same sensitivity factor that enters the first-layer gradient. At the null model, this factor therefore cancels from the local optimality condition, removing the neuron-specific effect of the downstream parametrization. The resulting zero-threshold condition depends only on the data-dependent score and can thus be calibrated through a single regularization parameter λ .

The following theorem turns the requirement that $W^{(1)} = \mathbf{0}$ be a local minimizer into a condition on λ . Consider the null parameter $\theta^0 = (W^{(1)} = \mathbf{0}, \hat{\tau})$, where $\hat{\tau}$ minimizes the loss subject to $W^{(1)} = \mathbf{0}$. Under this constraint, the network output is constant across observations, so that $\hat{\tau}$ yields the optimal constant fit under $\mathcal{L}_n$. For each first-layer neuron $k = 1, \dots, p_1$, let

$$\mathbf{g}_{0,k} = \nabla_{W_{k,\cdot}^{(1)}} \mathcal{L}_n(f_{\theta}(X); \mathbf{y}) \Big|_{\theta=\theta^0} \in \mathbb{R}^p \quad (6)$$

denote the corresponding first-layer gradient at the null model, and

$$\hat{\mathbf{r}}_0(\mathbf{y}) = \nabla_{\boldsymbol{\eta}} \mathcal{L}_n(\boldsymbol{\eta}; \mathbf{y}) \Big|_{\boldsymbol{\eta}=f_{\theta^0}(X)} \in \mathbb{R}^n \quad (7)$$

the gradient of the loss with respect to the network output, evaluated at the optimal constant fit.

Theorem 2 (Zero-thresholding function) Assume that $s_k(\boldsymbol{\theta}^0) > 0$ for all $k = 1, \dots, p_1$. The null parameter $\boldsymbol{\theta}^0 = (W^{(1)} = \mathbf{0}, \hat{\boldsymbol{\tau}})$ is a local minimizer of the cost function (P) if and only if

$$\lambda \geq \max_{k=1, \dots, p_1} \frac{\|\mathbf{g}_{0,k}\|_\infty}{s_k(\boldsymbol{\theta}^0)} = \|X^T \hat{\mathbf{r}}_0(\mathbf{y})\|_\infty =: \lambda_0(X, \mathbf{y}).$$

Proof At $W^{(1)} = \mathbf{0}$, the first-layer preactivation is identical across observations, $\mathbf{h}_i^{(1)} = \mathbf{b}^{(1)}$ for all $i = 1, \dots, n$. Since the downstream network therefore receives the same input for every observation, the first-layer sensitivity of neuron k is constant across the sample. Hence, for some scalar a_{0k} ,

$$\mathbf{a}_{\cdot k}(\boldsymbol{\theta}^0) = a_{0k} \mathbf{1}_n, \quad s_k(\boldsymbol{\theta}^0) = |a_{0k}|.$$

Let $\hat{\mathbf{r}}_0(\mathbf{y}) = \nabla_{\boldsymbol{\eta}} \mathcal{L}_n(\boldsymbol{\eta}; \mathbf{y})|_{\boldsymbol{\theta}=\boldsymbol{\theta}^0}$ denote the loss gradient at the optimal constant fit. By the chain rule, the gradient with respect to the k th row of the first-layer weights is

$$\mathbf{g}_{0,k} = X^\top (\mathbf{a}_{\cdot k}(\boldsymbol{\theta}^0) \odot \hat{\mathbf{r}}_0(\mathbf{y})) = a_{0k} X^\top \hat{\mathbf{r}}_0(\mathbf{y}).$$

Therefore,

$$\frac{\|\mathbf{g}_{0,k}\|_\infty}{s_k(\boldsymbol{\theta}^0)} = \frac{|a_{0k}| \|X^\top \hat{\mathbf{r}}_0(\mathbf{y})\|_\infty}{|a_{0k}|} = \|X^\top \hat{\mathbf{r}}_0(\mathbf{y})\|_\infty.$$

This quantity is identical for every first-layer neuron. Finally, the subgradient condition associated with the adaptive penalty at $W_{k,\cdot}^{(1)} = \mathbf{0}$ is $\|\mathbf{g}_{0,k}\|_\infty \leq \lambda s_k(\boldsymbol{\theta}^0)$ for $k = 1, \dots, p_1$. Since $s_k(\boldsymbol{\theta}^0) > 0$, these conditions hold simultaneously if and only if $\lambda \geq \max_{1 \leq k \leq p_1} \frac{\|\mathbf{g}_{0,k}\|_\infty}{s_k(\boldsymbol{\theta}^0)}$ yielding the provided statement. $\blacksquare$

The zero-thresholding function $\lambda_0(X, \mathbf{y})$ gives the critical regularization level at which $W^{(1)} = \mathbf{0}$ becomes locally optimal for the observed data $(X, \mathbf{y})$. To turn this data-dependent threshold into a calibrated regularization parameter, we evaluate its distribution in the setting where the null solution is appropriate, namely under H_0 . This null distribution provides the reference against which the regularization level is calibrated.

Definition 3 (Detection boundary) Let $\mathbf{Y}_0$ be a response vector generated under the pure-noise model. Define the random variable $\Lambda = \lambda_0(X, \mathbf{Y}_0)$, and let F_Λ denote its cumulative distribution function. Then, for a prescribed $\alpha \in (0, 1)$, the α -level detection boundary is

$$\lambda_\alpha^{\text{DB}} = F_\Lambda^{-1}(1 - \alpha). \quad (8)$$

The resulting $\lambda_\alpha^{\text{DB}}$ provides a fixed reference threshold calibrated under the null model. By construction, $\mathbb{P}_{H_0} \{\lambda_0(X, \mathbf{Y}_0) \leq \lambda_\alpha^{\text{DB}} \mid X\} = 1 - \alpha$. Combined with Theorem 2, this implies that $\lambda_\alpha^{\text{DB}}$ is the regularization level at which the null parameter satisfies the zero-threshold condition with probability $1 - \alpha$ under H_0 . This threshold is determined prior to network optimization and remains unchanged throughout training. Importantly, the downstream network parametrization does not enter its calibration: its contribution to the first-layer gradient has already been removed by the sensitivity scaling in Theorem 2. The calibration of λ therefore depends only on the distribution of the zero-thresholding function under H_0 .

Learning task	Zero-thresholding statistic $\lambda_0(X, \mathbf{y})$
Regression	$\frac{\ X^\top(\mathbf{y} - \bar{\mathbf{y}}\mathbf{1})\ _\infty / \sqrt{n}}{\ \mathbf{y} - \bar{\mathbf{y}}\mathbf{1}\ _2}$
Binary classification	$\frac{\ X^\top(\mathbf{y} - \bar{\mathbf{y}}\mathbf{1})\ _\infty / n}{\sqrt{\bar{\mathbf{y}}(1 - \bar{\mathbf{y}})}}$
Survival analysis	$\frac{\ n^{-1} \sum_{i=1}^n \delta_i(\mathbf{x}_i - \bar{\mathbf{x}}_{\mathcal{R}_i})\ _\infty}{2\{n^{-1} \sum_{i=1}^n \delta_i \log \mathcal{R}_i \}^{1/2}}$

Table 1: Pivotal zero-thresholding statistics for the three learning tasks considered in this work. For the survival analysis task, $\mathcal{R}_i = \{j : y_j \geq y_i\}$ and $\bar{\mathbf{x}}_{\mathcal{R}_i} = |\mathcal{R}_i|^{-1} \sum_{j \in \mathcal{R}_i} \mathbf{x}_j$

In general, such a null distribution may depend on unknown nuisance parameters, preventing the detection boundary from being calibrated without their estimation. The losses introduced in Section 3.3 are chosen precisely to avoid this difficulty. For each of the three learning tasks considered here, the corresponding zero-thresholding statistic $\Lambda = \lambda_0(X, \mathbf{Y}_0)$ is pivotal under H_0 , meaning that its null distribution is free of unknown nuisance parameters of $\mathbf{Y}_0$. Consequently, the detection boundary can be approximated by Monte Carlo simulation under any convenient choice of these nuisance parameters: repeated null samples $\mathbf{Y}_0^{(1)}, \dots, \mathbf{Y}_0^{(M)}$ are generated, the corresponding zero-thresholding statistics are computed, and $\lambda_\alpha^{\text{DB}}$ is estimated by their empirical $(1 - \alpha)$ -quantile.

Table 1 summarizes the resulting zero-thresholding statistic for regression, binary classification, and survival analysis. Further details on the pivotal construction are provided in Sardy et al. (2026), with the survival analysis setting developed specifically in van Cutsem and Sardy (2026).

4.2 Interpretation away from the null model

The calibration above is derived at the null model $W^{(1)} = \mathbf{0}$, where the sensitivity of each first-layer neuron is constant across observations and cancels exactly from the zero-threshold condition. We now consider the role of the same regularization parameter away from the null model.

For a general parameter $\boldsymbol{\theta}$ with $W^{(1)} \neq \mathbf{0}$, define the normalized sensitivity vector

$$\tilde{\mathbf{a}}_k(\boldsymbol{\theta}) = \frac{\mathbf{a}_k(\boldsymbol{\theta})}{s_k(\boldsymbol{\theta})}, \quad \|\tilde{\mathbf{a}}_k(\boldsymbol{\theta})\|_\infty = 1,$$

for every neuron such that $s_k(\boldsymbol{\theta}) > 0$. This decomposition separates two distinct components of the downstream sensitivity: $s_k(\boldsymbol{\theta})$ captures its overall magnitude, whereas $\tilde{\mathbf{a}}_k(\boldsymbol{\theta})$ preserves its variation across observations. In particular, the former measures how strongly neuron k can affect the network output, while the latter describes how this influence is distributed across the sample.

Let $\mathbf{r}(\boldsymbol{\theta}) = \nabla_{\boldsymbol{\eta}} \mathcal{L}_n(\boldsymbol{\eta}; \mathbf{y})|_{\boldsymbol{\eta}=f_{\boldsymbol{\theta}}(X)}$ denote the loss gradient with respect to the network output. By the chain rule, the gradient associated with the k th first-layer neuron satisfies

$$\mathbf{g}_k(\boldsymbol{\theta}) = X^\top (\mathbf{a}_k(\boldsymbol{\theta}) \odot \mathbf{r}(\boldsymbol{\theta})) = s_k(\boldsymbol{\theta}) X^\top (\tilde{\mathbf{a}}_k(\boldsymbol{\theta}) \odot \mathbf{r}(\boldsymbol{\theta})).$$

Consequently, for each first-layer connection (k, j) , the corresponding threshold condition can be written as

$$|g_{kj}(\boldsymbol{\theta})| \leq \lambda s_k(\boldsymbol{\theta}) \iff \left| X_j^\top (\tilde{\mathbf{a}}_k(\boldsymbol{\theta}) \odot \mathbf{r}(\boldsymbol{\theta})) \right| \leq \lambda.$$

Hence, the adaptive penalty removes the overall magnitude $s_k(\boldsymbol{\theta})$ of the downstream sensitivity from the thresholding condition, while retaining its observation-specific structure through $\tilde{\mathbf{a}}_k(\boldsymbol{\theta})$. The regularization parameter λ therefore acts on a normalized first-layer score and can be shared across all first-layer neurons.

On a normalized network, $s_k = 1$ and the condition simply reads $|g_{kj}(\boldsymbol{\theta})| \leq \lambda$: the zero-thresholding condition of a lasso at level λ , the one calibrated at the null.

5 Optimization

5.1 A Reversed Warm Path

While the ultimate goal is to solve (P), the calibrated level $\lambda_\alpha^{\text{DB}}$ should not in general be imposed directly at initialization. The reason is that $\lambda_\alpha^{\text{DB}}$ is calibrated from the first-order behavior of the network at the null model. A relevant variable may nevertheless have little or no marginal association with the response when its effect is purely nonlinear. Consequently, its relevance may not be visible from the first-order score at the null model. Starting the optimization directly at $\lambda_\alpha^{\text{DB}}$ may then drive the first-layer weights to zero before the network has learned the nonlinear representation through which that variable becomes informative.

We therefore take the conservative approach of solving for a sequence of λ 's in the spirit of simulated annealing. Namely, we propose solving (P) using an increasing grid

$$0 < \lambda^{(1)} < \lambda^{(2)} < \dots < \lambda^{(M)} = \lambda_\alpha^{\text{DB}}.$$

Taking as initial parameter values the solution corresponding to the previous $\lambda^{(i)}$ leads to a sequence of sparser approximating solutions until solving for $\lambda_\alpha^{\text{DB}}$ at the last step. Along this path, the normalized sensitivities $\tilde{\mathbf{a}}_k(\boldsymbol{\theta})$ evolve with the learned representation. Hence, once nonlinear structure has been learned, the adaptive first-layer score $X_j^\top (\tilde{\mathbf{a}}_k(\boldsymbol{\theta}) \odot \mathbf{r}(\boldsymbol{\theta}))$ can reflect information that is absent from the marginal score at the null model. Increasing λ then progressively removes first-layer connections that are no longer supported by these learned, observation-specific scores.

5.2 One Iteration

Let φ denote all the parameters other than $W^{(1)}$ (every bias and $W^{(2)}, \dots, W^{(L)}$), let $\mathbf{g}_k = \nabla_{W_{k,\cdot}^{(1)}} \ell$, and let $\mathcal{S}_\tau(x) = \text{sign}(x) \max(|x| - \tau, 0)$ denote soft-thresholding. Within a phase at level $\lambda = \lambda^{(m)}$, each iteration does three things.

1. Normalize. Every neuron is rescaled by (4) with $c = s_k(\boldsymbol{\theta})$, so that $s_k = 1$. This changes neither the fit nor J_λ . It matters for two reasons. First, on the invariant coefficients $s_k W_{k,\cdot}^{(1)}$, a gradient step of size δ is a step of size δs_k^2 : without normalization, a neuron whose sensitivity shrinks would freeze before its weights reach zero, leaving spurious variables in

the model. Second, gradient steps are not invariant to rescaling, so the trajectory would otherwise depend on an arbitrary scale. Since gradients and Adam’s moments rescale with the weights, they are rescaled along: divided by c (first moment) or c^2 (second moment) for $W_{k,\cdot}^{(1)}$ and $b_k^{(1)}$, multiplied likewise for $W_{\cdot,k}^{(2)}$.

2. Proximal step on the first layer. With $s_k = 1$, the penalty on $W^{(1)}$ is $\lambda \|W^{(1)}\|_1$, handled by a proximal step in the metric of Adam (Kingma and Ba, 2017; Yun et al., 2021):

$$W_{kj}^{(1)} \leftarrow \mathcal{S}_{D_{kj}\lambda}(W_{kj}^{(1)} - D_{kj} g_{kj}), \quad D_{kj} = \frac{\delta}{\sqrt{\hat{v}_{kj}} + \varepsilon}, \quad (9)$$

where $\hat{v}_{kj}$ is the moving average of g_{kj}^2 . A zero weight stays at zero if and only if $|g_{kj}| \leq \lambda$: the step size D_{kj} cancels, and the test is exactly the calibrated one. No momentum is used on $W^{(1)}$: with momentum, the test would bear on past gradients instead of the current one.

3. Adam step on the other parameters. The penalty depends on φ through the sensitivities, so φ follows

$$\nabla_{\varphi} J_{\lambda} = \nabla_{\varphi} \ell + \sum_{k=1}^{p_1} \mu_k \nabla_{\varphi} s_k, \quad \mu_k = \lambda \|W_{k,\cdot}^{(1)}\|_1. \quad (10)$$

The second term is essential. Without it, the only fixed points have $W^{(1)} = \mathbf{0}$: no variable could ever be selected (Proposition 5 in Appendix A). In a one-neuron example whose solution is $u^* = 1$, dropping it makes the iterations settle at $u \approx 1.75$. In (C), this term is the gradient of the constraint $s_k = 1$ weighted by its Lagrange multiplier, and the multiplier is known in closed form, μ_k .

A phase stops once J_{λ} has changed, relatively, by less than a tolerance over the last 50 iterations.

5.3 What Is Guaranteed

Proposition 4 (a) Fixed points. *A normalized network (with zero first moment on φ) is left unchanged by an iteration if and only if $\nabla_{\varphi} J_{\lambda} = \mathbf{0}$ and, for every k and j , $g_{kj} = -\lambda \text{sign } W_{kj}^{(1)}$ when $W_{kj}^{(1)} \neq 0$ and $|g_{kj}| \leq \lambda$ otherwise: the first-order conditions of (C).*

(b) Calibration preserved. *The normalized null network $\theta^0 = (W^{(1)} = \mathbf{0}, \hat{\tau})$ is left unchanged if and only if $\lambda \geq \lambda_0(X, \mathbf{y})$.*

By (b), the iteration applies exactly the test of Theorem 2 for which $\lambda_{\alpha}^{\text{DB}}$ is calibrated. The proof is in Appendix A, and holds where the activation patterns are locally constant. No convergence result is claimed: J_{λ} is not continuous, as s_k jumps when an activation flips, so the guarantees of proximal adaptive methods (Yun et al., 2021) do not apply, and Adam itself may fail to converge (Reddi et al., 2018).

Algorithm 1 Fit along the warm path

Require: X (centred, unit second moment), $\mathbf{y}$, level α , number of phases M

- 1: $\lambda_\alpha^{\text{DB}} \leftarrow (1 - \alpha)$ -quantile of $\|X^\top \hat{\mathbf{r}}_0(\mathbf{Y}_0)\|_\infty$ over Monte Carlo draws $\mathbf{Y}_0$ under H_0
 - 2: **for** $\lambda = \lambda^{(1)} < \dots < \lambda^{(M)} = \lambda_\alpha^{\text{DB}}$ (warm start) **do**
 - 3: **repeat**
 - 4: compute $\nabla \ell$, the sensitivities s_k and $\nabla_\varphi J_\lambda$ by (10)
 - 5: normalize: rescale every neuron by s_k , with its gradients and moments
 - 6: $W^{(1)} \leftarrow$ proximal step (9); $\varphi \leftarrow \text{Adam}(\varphi, \nabla_\varphi J_\lambda)$
 - 7: **until** J_λ has stabilized
 - 8: **end for**
 - 9: **return** $\hat{S} = \{j : \hat{W}_{\cdot,j}^{(1)} \neq \mathbf{0}\}$
-

5.4 Implementation

Sensitivities. Each η_i depends on $\mathbf{x}_i$ only, so $\partial \ell / \partial \mathbf{h}_i^{(1)} = r_i \mathbf{a}_i$ with $r_i = \partial \ell / \partial \eta_i$: the sensitivities come with the backward pass of the loss, or, where $r_i = 0$, from (2) at the cost of one forward pass of the deeper layers.

The extra gradient term. With the activation patterns frozen, the network at observation i is the linear map $\mathbf{u} \mapsto \mathbf{a}_i^\top \mathbf{u}$. Feeding $\mathbf{u} = \mu_k \text{sign}(a_{i_k^* k}) \mathbf{e}_k$ at the observation i_k^* where s_k is attained gives the output $\sum_k \mu_k s_k$, whose gradient is the second term of (10). It costs a forward and a backward pass of the deeper layers on p_1 rows, so an iteration costs about one epoch of a standard network.

Active set. A first-layer column that is zero when a phase starts is left out of the computation, unless it fails the test $\max_k |g_{kj}| \leq \lambda$. At the end of the last phase, the left-out columns are tested at $\lambda_\alpha^{\text{DB}}$, and any that fail are put back and the phase is run again. The final selection is therefore the one the full problem would give, as in the active-set strategy of glmnet (Friedman et al., 2010).

5.5 Optional Refitting on the Selected Model

After completion of the regularization path, the selected support is determined from the nonzero columns of $\hat{W}^{(1)}$. If desired, the network can subsequently be reduced to this selected set of input features, together with the removal of any hidden neurons that have become inactive. The dimensions of $\hat{W}^{(2)}$ are adjusted accordingly. An optional final training phase can then be performed on this reduced architecture without regularization so as to reduce any estimation bias introduced by the penalized fit.

6 Experiments

Appendix A. Proofs

Throughout, the activation patterns are assumed constant near the point considered, so that ℓ and the s_k are differentiable there. Then s_k does not depend on $W^{(1)}$ nor on the biases, depends on $W^{(2)}$ only through its column $W_{\cdot,k}^{(2)}$, and is linear in that column.

An identity. Since ℓ is unchanged by (4), differentiating in c at $c = 1$ gives, for every neuron k ,

$$\langle W_{k,\cdot}^{(1)}, \mathbf{g}_k \rangle + b_k^{(1)} \partial_{b_k^{(1)}} \ell - \langle W_{\cdot,k}^{(2)}, \nabla_{W_{\cdot,k}^{(2)}} \ell \rangle = 0. \quad (11)$$

Proposition 5 *Let $\lambda > 0$ and $s_k > 0$ for every k . If $\nabla_\varphi \ell = \mathbf{0}$ and the first-layer conditions of Proposition 4(a) hold, then $W^{(1)} = \mathbf{0}$.*

Proof By (11), $\nabla_\varphi \ell = \mathbf{0}$ gives $\langle W_{k,\cdot}^{(1)}, \mathbf{g}_k \rangle = 0$, whereas the first-layer conditions give $\langle W_{k,\cdot}^{(1)}, \mathbf{g}_k \rangle = -\lambda s_k \|W_{k,\cdot}^{(1)}\|_1$. Hence $W_{k,\cdot}^{(1)} = \mathbf{0}$ for every k . $\blacksquare$

Proof of Proposition 4. (a) On a normalized network, the normalization does nothing. For scalars and $D > 0$, $x = \mathcal{S}_{D\lambda}(x - Dg)$ holds if and only if $g = -\lambda \text{sign } x$ when $x \neq 0$ and $|g| \leq \lambda$ when $x = 0$, which gives the first-layer conditions. On φ , the first moment becomes $m' = \beta_1 m + (1 - \beta_1) \nabla_\varphi J_\lambda$ and the step is proportional to m' ; with $m = \mathbf{0}$, nothing moves if and only if $\nabla_\varphi J_\lambda = \mathbf{0}$.

(b) At $W^{(1)} = \mathbf{0}$, every observation reaches the deeper layers with the same input $\mathbf{b}^{(1)}$, so $a_{ik} = a_{0k}$ for all i , with $|a_{0k}| = s_k = 1$. As in the proof of Theorem 2, $\mathbf{g}_k = a_{0k} X^\top \hat{\mathbf{r}}_0(\mathbf{y})$, so $W^{(1)}$ stays at zero if and only if $\|X^\top \hat{\mathbf{r}}_0(\mathbf{y})\|_\infty \leq \lambda$. On φ , $\mu_k = 0$, and since $\partial \eta_i / \partial \varphi$ is the same for every i , $\nabla_\varphi J_\lambda = (\sum_i \hat{r}_{0i}) \partial \eta_1 / \partial \varphi$. The sum vanishes: it is the derivative of $c \mapsto \mathcal{L}_n(c\mathbf{1}; \mathbf{y})$ at its minimizer $\hat{\eta}$, or, for the Cox model, whose loss is invariant to adding a constant to $\boldsymbol{\eta}$, it vanishes identically.

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